

Cumulative Distribution Function (CDF)

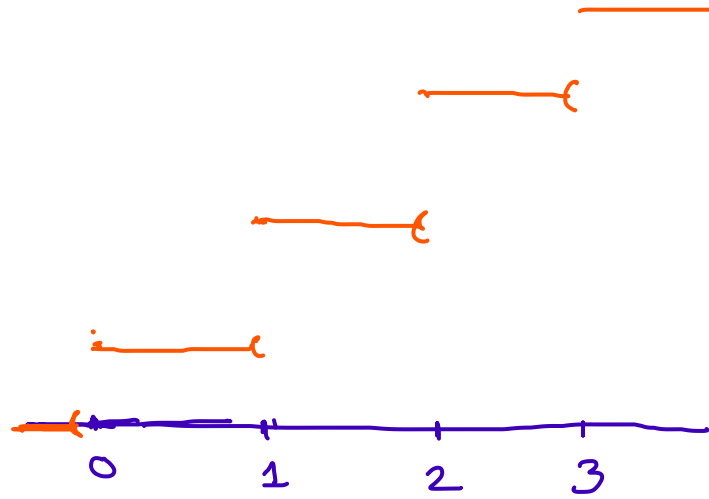
$F_x: \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_x(x) = P[X \leq x], \quad x \in \mathbb{R}$$

$$= P[-\infty < X \leq x]$$

$$= P\{\omega: -\infty < X(\omega) \leq x\}$$

$$F_x(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{8}, & 0 \leq x < 1 \\ \frac{4}{8}, & 1 \leq x < 2 \\ \frac{7}{8}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$



Properties of the CDF:

(i) $F(+\infty) = \lim_{x \rightarrow +\infty} F(x) = 1$

$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0$$

(ii) F is a non-decreasing function of x .

i.e., $x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$

(iii) F is right continuous function.

$$\lim_{h \rightarrow 0} F(x+h) = F(x).$$

$$F_X(x) = P(X \leq x) \\ = \sum_{x_i' \leq x} p_X(x_i')$$

$$\therefore p_X(x_i) = F_X(x_i) - F_X(x_{i-1})$$

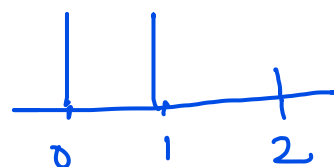
Ex: A computer store contains 10 computers of which 3 are defective. A customer buys 2 at a random. Let x denote no. of defectives in the purchase.

$x: 0, 1, 2$

$$p_X(0) = P(X=0) = \frac{\binom{7}{2}}{\binom{10}{2}} = \frac{21}{45} = \frac{7}{15}$$

$$p_X(1) = P(X=1) = \frac{\binom{7}{1} \binom{3}{1}}{\binom{10}{2}} = \frac{21}{45} = \frac{7}{15}$$

$$p_X(2) = P(X=2) = \frac{\binom{3}{2}}{\binom{10}{2}} = \frac{3}{45} = \frac{1}{15}$$



$$F_X(x) = \begin{cases} 0, & x < 0 \\ \frac{7}{15}, & 0 \leq x < 1 \\ \frac{14}{15}, & 1 \leq x < 2 \\ 1, & x \geq 2. \end{cases}$$

PMF
↓
CDF

$[0, 15]$

A r.v. X is said to be continuous if its cdf F_X is absolutely continuous i.e., $\exists$ a non-negative function $f_X(x)$ such that $F_X(x) = \int_{-\infty}^x f_X(t) dt$ for all $x \in \mathbb{R}$.

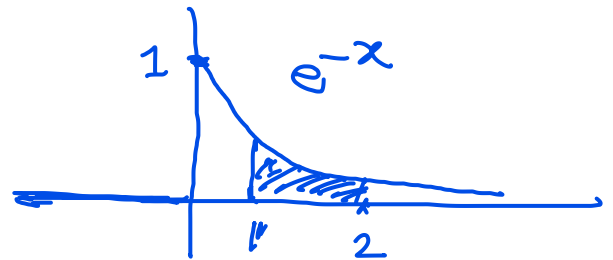
The function $f_X(x)$ is called the probability density function of X . It satisfies

$$(i) f_X(x) \geq 0 \quad \forall x \in \mathbb{R}$$

$$(ii) \int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$(iii) \int_a^b f_X(x) dx = P(a < X \leq b)$$

Ex: (i) $f_X(x) = e^{-x}, x \geq 0$
 $= 0$, otherwise



$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} e^{-x} dx = [-e^{-x}]_0^{\infty} = 1$$

$$\begin{aligned} P(1 \leq X \leq 2) &= \int_1^2 f(x) dx \\ &= \int_1^2 e^{-x} dx \\ &= [-e^{-x}]_1^2 \\ &= e^{-1} - e^{-2} \end{aligned}$$

$$P(X > 5) = \int_5^{\infty} e^{-x} dx$$

PDF $\Rightarrow$ CDF

$f_X(x)$

$$F_X(x) = \underline{P}(-\infty < X \leq x) \\ = \int_{-\infty}^x f_X(t) dt$$

Ex: $f_X(x) = e^{-x}, x \geq 0$
 $= 0, \text{ otherwise}$

If $x < 0$, $F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt = 0$

if $x \geq 0$, $F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt = \int_{-\infty}^0 f_X(t) dt + \int_0^x f_X(t) dt$
 $= \int_0^x e^{-t} dt$
 $= [-e^{-t}]_0^x$
 $= 1 - e^{-x}$

$$F_X(x) = 0, x < 0$$
$$= 1 - e^{-x}, x \geq 0$$

